

StockSense Workspace Setup

Tool & Development Environment Verification Checklist

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1. ENVIRONMENT VERIFICATION SUMMARY

TOOL / SOFTWARE	INSTALLED VERSION	STATUS	VERIFICATION METHOD
Git	2.54.0 (Windows)	Verified	<code>git --version</code> executed successfully
Python	3.13.x	Verified	Executed <code>py --version</code>
Node.js	v24.14.0	Verified	Executed <code>node --version</code>
Docker Engine	29.7.2	Verified	Executed <code>docker run hello-world</code>
VS Code	Latest Stable	Verified	Verified PlatformIO IDE Extension installed
PlatformIO	ESP32 Core	Verified	Initialized sample ESP32 test project
KiCad	PCB EDA Suite	Verified	Application installed for schematic capture
Figma	Desktop / Web UI	Verified	Account created & client accessible
MQTT Explorer	Desktop Client	Verified	Utility installed for MQTT debugging

2. ENVIRONMENT VERIFICATION TESTS

Test A: Docker Engine & Container Daemon Test

Command: `docker run hello-world`

Output Verification: Container pulled successfully from Docker Hub and executed daemon test:

```
Hello from Docker!  
This message shows that your installation appears to be working correctly.
```

Test B: Python Analytics & Data Science Stack

Installation Command: `py -m pip install pandas scikit-learn statsmodels paho-mqtt`

Verification Execution Script:

```
py -c "import pandas, sklearn, statsmodels, paho.mqtt; print('All packages installed  
successfully!')"
```

Result: Output printed `All packages installed successfully!` without missing dependencies.

Test C: Firmware Workspace & PlatformIO Setup

